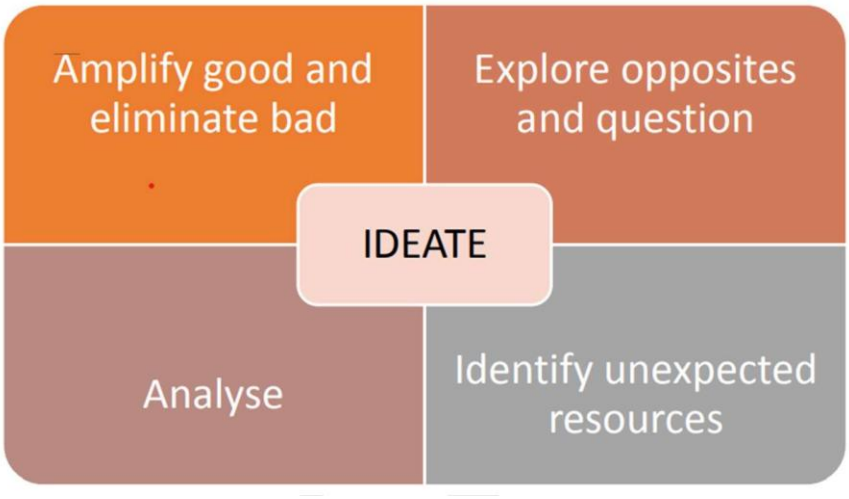
IDEATION

INTRODUCTION

The third component of design thinking process is the most interesting and perhaps, the most rigorous as well. In this section, called Ideate, a design thinker is supposed to bring to the table as many ideas as possible. While brainstorming for ideas, it is not checked whether the idea is possible, feasible, and viable or not. The only task of thinkers is to think of as many ideas as possible for them.



In this process, design thinkers also resort to the use of boards, sticky notes, sketching, chart papers, mind maps, etc. We will look at mind maps later in this section. Design thinkers also build on the ideas of other design thinkers. All solutions suggested by design thinkers are brought to the table and thought over.

TYPES OF THINKING

There are four types of "thinking skills": convergent or analytical thinking, divergent thinking, critical thinking and creative thinking. We use these skills to help us understand the world around us, think critically, solve problems, make logical choices and develop our own values and beliefs.

The 4 types of thinking skills are:

1. Divergent Thinking
2. Convergent Analytical Thinking
3. Critical Thinking and
4. Creative Thinking

1. Divergent Thinking

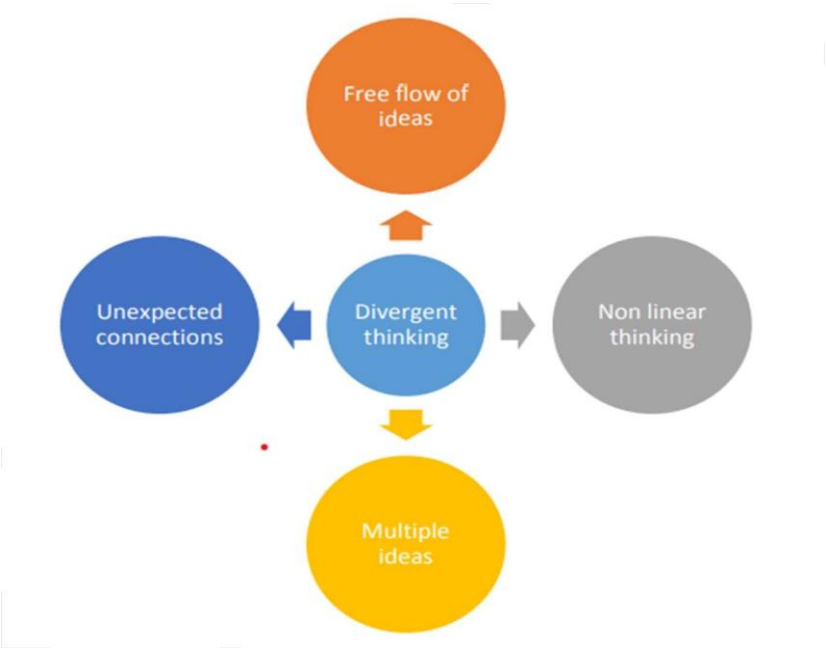
The term 'Divergent Thinking' was first coined by J. P. Guilford in 1956. Divergent thinking involves coming up to solutions, paths forward or new ideas when there is no single correct answer.

Questions like “should I study to become a doctor or a lawyer?” may not have a simple answer. You might be good at both, and both options might bring you happiness and a good life. So, which option should you choose?

To come up with solutions to questions without clear answers, you need to break down the possibilities and analyze each part. You might create a pros and cons list, a Venn diagram or a table to lay out your options and consider each one in turn.

The main features of divergent thinking are:

- It is a free-flowing chain of ideas.
- It happens in a non-linear manner, i.e., it does not follow any particular sequence of thinking. Moreover, multiple ideas can emerge at the same time, rather than one idea coming up only after the other has occurred.
- Non-linearity also means that multiple solutions are thought of and explored at the same time. This happens in a very short amount of time and unexpected connections are developed between the ideas.



Case Study

Problem Statement: The process of knowledge transfer is a huge problem for the organization. Let’s call our organization ‘DT’. DT wants to eliminate the overhead of shelling out extra money and investing time for transferring knowledge to its new employees. The problem statement at hand is “Knowledge transfer adds to the cost of the company”. Let’s think of ways to eliminate or at least, reduce the cost to the company.

Solution: Following can be some of the possible and even not-so-possible solutions.

- DT can eliminate the process of knowledge transfer.

- DT can conduct classroom sessions for knowledge transfer, where many new employees can be seated and just one instructor can deliver sessions to many employees at once. This will reduce the cost as the number of paid instructors required will be less.

- DT can come up with a document for knowledge transfer and can mail it to every new employee. The employees can go through the document and hence, can self-help for knowledge transfer.

- DT can ask the employees to search for material online to gain knowledge of new tools and processes, which are currently in use in the industry.

- DT can hire only those employees who have adequate knowhow of tools and techniques that DT works on. This will eliminate the need of knowledge transfer.

There may be many other solutions that may come to your mind. Write them down on a sheet of paper. Here, we won't focus on whether a solution is possible, feasible or viable. We just need to bring ideas to the table, no matter how absurd they may sound. This is called the process of divergent thinking, where a thinker is free to move or flow in any direction.

One great example of divergent thinking in action is in the early days of Twitter. Twitter took a Minimum Viable Product (MVP) approach to their application. They created an online service without a clear practical application or market, launched it to see how people would use it, then refined it. This doesn't mean that launching something and then figuring out what the market is for it is a bullet-proof strategy. In Twitter's case, it worked. In most cases, it doesn't. You just don't hear about the failures.

2. Convergent Analytical Thinking

Convergent thinking is the exact opposite of convergent thinking. The term 'Convergent Thinking' was coined by Joy Paul Guilford in 1956. It is the process of coming up with the best answer to a question using our memory, resources around us, or logic.

This thinking skill does not require significant creativity or lateral thinking strategies. It is not the best for solving problems that are complex or require thinking out of the box. Instead, it uses very straightforward thought processes. A convergent thinker simply needs to apply already established procedures and memory recall to reach the 'correct' answer.

Convergent thinking is very commonly used for standardized and multiple choice tests. These sorts of tests simply assess our knowledge and ability to apply knowledge to simple and logical situations.

The key elements required to be a skilled convergent thinker are: speed, accuracy and logic.

Case Study

Problem Statement: The process of knowledge transfer is a huge problem for the organization. Let's call our organization 'DT'. DT wants to eliminate the overhead of shelling out extra money and investing time for transferring knowledge to its new employees. The problem statement at hand is "Knowledge transfer adds to the cost of the company". Let's think of ways to eliminate or at least, reduce the cost to the company.

We got the following ideas in the divergent thinking exercise.

- Elimination of knowledge transfer program.
- Having a single instructor for knowledge transfer program in a classroom session.
- Preparing a document for knowledge transfer program.
- Making it mandatory for employees to search for knowledge resources online.
- Hiring only those employees who are experienced enough and who don't need knowledge transfer.

ow, looking at the five ideas, it can be easily said that option 1 is not feasible. Every employee does not have an idea of a company's tools and techniques and hence, cannot be expected to survive without knowledge transfer.

For the same reason, option 5 is also not acceptable. The best practices of a company are seldom known to new employees and taking an assumption about an employee's knowledge level is a huge mistake. It is considered to be a good HR practice to have knowledge transfer session for new employees.

If we go by option 4, we are not assured of the pace at which learning will happen for the new employees. Each employee can take variable amount of time to grasp the concepts. The time taken to search materials online and read them is an overhead in itself and it cannot be monitored.

Hence, the two better options that remain are option 2 and option 3. However, one cannot correctly estimate the effectiveness of a document for knowledge transfer. It is similar to reading materials online. Hence, the best option available is to have an instructor teaching employee in a classroom program.

Although, the employees won't get personal attention at times, yet by maintaining a fine balance between the strength of the batch and the length of class, this can be the best option to reduce cost and overhead. The reduction in the number of instructors will lead to less expenditure for DT and at the same time, the effectiveness of a paid instructor will remain, making the process of knowledge transfer as effective as before.

This is how convergent thinking comes into picture.

Divergent vs. Convergent Thinking

Here's an example of the same problem framed for divergent and convergent thinking:

Divergent Example

I live four miles from work. My car gets 30 MPG. I want to use less fuel in my commute for financial and conservation reasons. Money is no object. *What options do I have to reduce my fuel consumption?*

Convergent Example

I live four miles from work. My car gets 30 MPG. I want to use less fuel in my commute for financial and conservation reasons. Money is no object. *Find the three best replacement vehicles for my car*.

The Difference

The problem is the same, but the questions change slightly. The convergent example asks for a vehicle, whereas the divergent example doesn't rule out options like moving closer to work, telecommuting, walking, carpooling, or taking public transportation.

Both examples will produce valuable results. The convergent example may be driven by other issues. Perhaps my current car was totalled and I only have a weekend to solve the problem. The divergent example may take more time to investigate, but you may discover an option that is completely different from what the user has asked you to do. As designers, we need to use both methods of thinking to best serve our clients.

3. Critical Thinking

Critical thinking skills involve analyzing something in order to form a judgement about it. Developing your critical thinking skills improves your problem solving skills, boosts your data-driven decision making ability, and gives you a methodology to tackle complex problems. Good critical thinkers are comfortable with ambiguity and are willing to challenge their hypotheses in order to come to the best conclusions.

A critical thinker does not take the assumptions of a topic for granted. Instead, the critical thinking involves 'critiquing' what you are viewing using your available intellectual knowledge.

People who think critically can use three processes to develop critical insights on a topic: ***deduction, induction and abduction***.

Deduction includes the critical thinking skills that involve drawing conclusions based on the facts at hand. You have all the facts available to you to come to a clear and unambiguous conclusion about a topic. For example, a doctor does blood tests to determine if someone has a virus. The blood tests come back positive, so we can deduce that you definitely have that virus. Deduction is a great skill to use if you want to solve problems.

Induction includes the critical thinking skills that involve drawing conclusions based on a generalization. You don't have all the exact information at hand. However, you think critically and realize are aware of patterns, clues and a methodology that can help you induce the answer. For example, you come to the doctor exhibiting a fever, sneezing and coughing. The doctor doesn't do tests, but they induce that you probably have influenza because your symptoms are characteristic of someone with the flu.

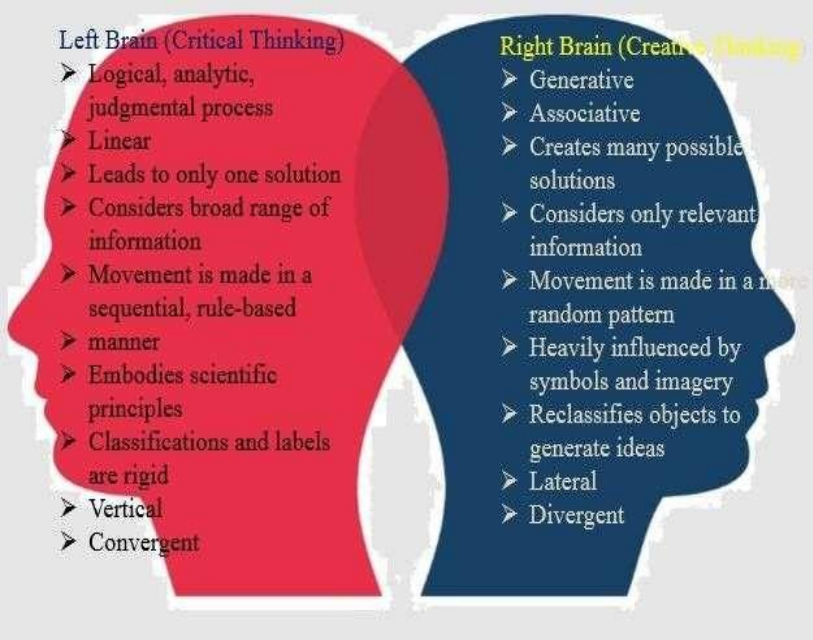
Abduction includes the critical thinking skills that involve coming to a conclusion that is the most likely or logical based on the small amount of knowledge that you have. You can't be sure of the answer, but you can think critically and make an educated guess. For example, you may see that a cat is on the roof. The most logical answer is that the cat got up there by climbing a nearby tree and jumping from it to the roof, but you can't be sure.

4. Creative Thinking

Creative thinking involves thinking about a topic in unusual, unconventional and alternative ways to generate new ideas about an established topic. A creative thinker will try to address an issue from a perspective that hasn't been used before.

While creative thinking may appear illogical, it is in fact a great driver of human development. Creative thinkers identify gaps in marketplaces or new, easier, faster and better ways of doing things. When a creative thinker comes up with a great new way of approaching an issue, their new method can become the new orthodoxy.

Creative thinkers aren't afraid to take risks and try new ideas. In fact, this ability to develop, test, and implement original solutions makes them a valuable asset to just about any workplace.



IDEATION METHODS

BRAINSTORMING:

Brainstorming is the most common method used by design teams for generating ideas. This method was developed by Alex Osborn to stimulate creative magazine advertisements, but it has been widely adopted in other areas such as design. Brainstorming is a carefully orchestrated process. It makes use of the broad experience and knowledge of groups of individuals. A well-done brainstorming session is an enthusiastic session of rapid, free-flowing ideas.

To achieve a good brainstorming session, it is important to carefully define the problem at the start. It is also necessary to allow a short period for individuals to think through the problem quietly and on their own before starting the group process. Participants in brainstorming sessions react to ideas they hear from others by recalling their own thoughts about the same concepts. This action of redirecting a stream of thought uncovers new possibilities in the affected team member. Some new ideas may come to mind by adding detail to a recently voiced idea or taking it in different, but related, directions. This building upon others ideas is known as piggy-backing or scaffolding, and it is an indicator of a well-functioning brainstorming

session. It has been found that the first 10 or so ideas will not be the freshest and creative, so it is critical to get at least 30 to 40 ideas from your session.

An important attribute of this method is that brainstorming creates a large number of ideas, some of which will be creative. The evaluation of your ideas should be done at a meeting on a day soon after the brainstorming session. This removes any fear that criticism or evaluation is coming soon and keeps the brainstorming meeting looser. Also, making the evaluation on the day after the idea generation session allows incubation time for more ideas to generate and time for reflection on what was proposed. The evaluation meeting should begin by adding to the original list any new ideas realized by the team members after the incubation period. Then the team evaluates each of the ideas. Hopefully, some of the wild ideas can be converted to realistic solutions.

One way to help the brainstorming process is to break up the normal thought pattern by using a checklist to help develop new ideas. The originator of brainstorming proposed such a list, which Eberle modified into the acrostic SCAMPER (Table 3.1).

Generally, the SCAMPER checklist is used as a stimulant when the flow of ideas begins to fall off during the brainstorming activity. The questions in the SCAMPER checklist are applied to the problem in the following way:

- Read aloud the first SCAMPER question.
- Write down ideas or sketch ideas that are stimulated by the question.
- Rephrase the question and apply it to the other aspects of the problem.
- Continue applying the questions until the ideas cease to flow.

Because the SCAMPER questions are generalized, they sometimes will not apply to a specific technical problem. Therefore, if a question fails to evoke ideas, move on quickly to the next question.

Table 3.1

SCAMPER Checklist to Aid in Brainstorming

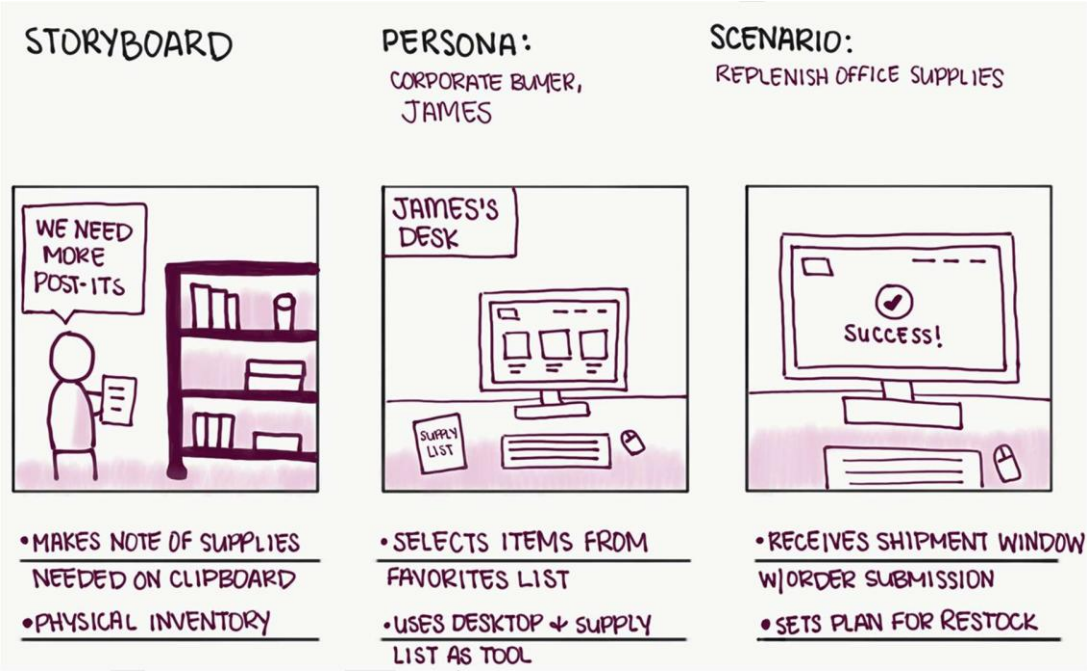
Proposed Change	Description
Substitute	What if used in a different material, process, person, power source, place, or approach?
Combine	Could I combine units, purposes, or ideas?
Adapt	What else is like this? What other idea does it suggest? Does the past offer a parallel? What can I copy?
Modify, magnify, minify	Could I add a new twist? Could I change the meaning, color, motion, form, or shape? Could I add something? Make stronger, higher, longer, thicker? Could I subtract something?
Put to other uses	Are there new ways to use this as is? If I modify it, does it have other uses?
Eliminate	Can I remove a part, function, person without affecting outcome?
Rearrange, reverse	Could I interchange components? Could I use a different layout or sequence? What if I transpose cause and effect? Could I transpose positive and negative? What if I turn it backward, upside down, or inside out?

STORYBOARDING

Storyboarding is one of the most valuable stages of the design thinking process. Storyboards are a visual representation of the user's journey. Before actual products can be designed, built, and launched, collaborators first need to research their target user's problems and then distill that down into a story. What is the user's starting point? What do they want to achieve? It can also help organizations save money, because they'll iron out disagreements even before investing in prototypes.

Storyboarding is most commonly done during the discovery phase for a new product to help align stakeholders around user outcomes. But it can also be important undertaking when developing an existing product.

A storyboard doesn't have to be complicated or high-fidelity. Simple visuals and a basic, but specific scenario will be memorable for your team and stakeholders.



Components of a Storyboard

There are always 3 common storyboard elements, regardless of form: a specific scenario, visuals, and corresponding captions.

1. Scenario

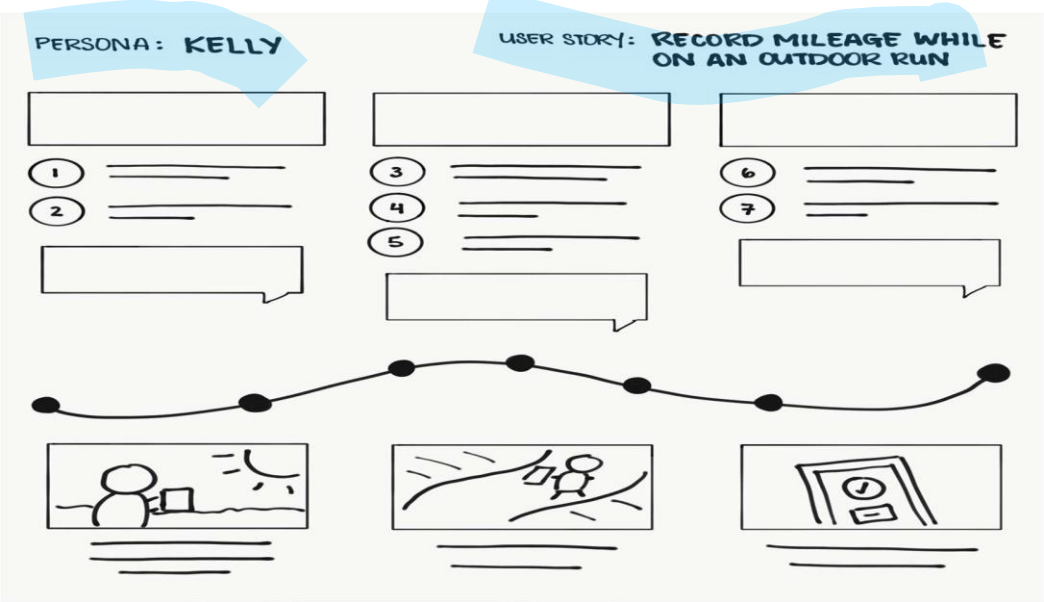
Storyboards are based on a scenario or a user story. The persona or role that corresponds to that scenario is clearly specified at the top of the storyboard. A short text description of the scenario is also included. The description of the scenario or story is clear enough that a team member or stakeholder could understand what is depicted before looking at the visuals. For example: *Corporate buyer, James, needs to replenish office supplies.*

2. Visuals

Each step in the scenario is represented visually in a sequence. The steps can be sketches, illustrations, or photos. Depending on the purpose of the storyboard and on its audience, these images can be quick, low-fidelity drawings or elaborate, high-fidelity artifacts. Images include details relevant to the story, such as what the user's environment looks like, speech bubbles with quotes from the user, or a sketch of the screen that the user is interacting with.

3. Captions

Each visual has a corresponding caption. The caption describes the user's actions, environment, emotional state, device, and so on. Because the image is the primary content in a storyboard, captions are concise and don't typically exceed two bullet points.



Coupling a journey map (top) with a storyboard (bottom) can provide insight to your story.

Process: How to Create a Storyboard

Creating a storyboard can feel like a daunting task. However, remember they are most valuable as a low-fidelity artifact and don't need to take a long time to create.

Effective storyboarding follows 6 key high-level steps:

1. Gather your data

First, determine which data will be used in your storyboard — user interviews, usability tests, or site metrics.

It is possible to do a storyboard without real data if you haven't collected data yet or you want to use storyboards as a form of ideation.

2. Choose your fidelity level

Keep in mind the goal and the audience of your artifact.

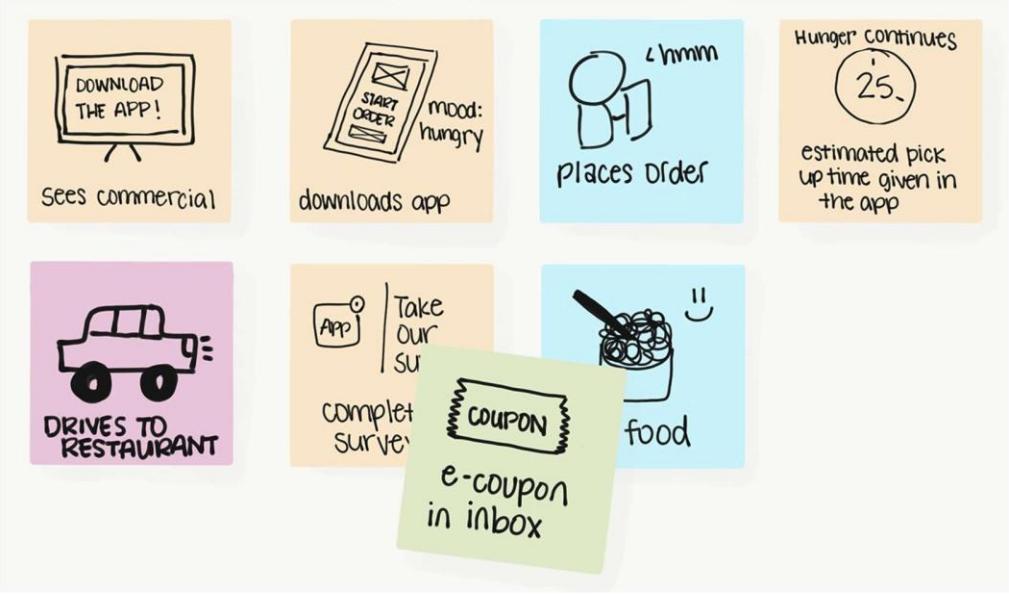
Use sketches to quickly draw a sequence or communicate a scene to your team during a brainstorming meeting. In such ideation meetings, you can even create storyboards collaboratively with sticky notes, to get each team member's perspective. Start by having a discussion of the timeline and steps the user will take. As you discuss, draw each step on a

sticky note and place on a whiteboard or a wall. You can designate one person to draw or multiple people can draw, as long as the discussion is happening as a group. Focus on a single step at a time to keep the discussion global and avoid splitting the group into multiple subgroups with different conversations that must be then consolidated. Having multiple team members present during this process will bring up ideas that a single role won't necessarily think about. For example, a marketing or business-focused team member may contribute a step (like a trigger for a marketing email or coupon, for example) that the rest of the team didn't have on the radar. This method also gives you the flexibility to change the sequence of events by rearranging sticky notes (especially if you're still adjusting the story timeline) without having to redraw the entire storyboard. The goal is to form a shared understanding, rather than a polished, refined artifact.

If you've recorded a usability test and are creating a storyboard to distill the information down, use photos or video stills. These types of visuals maximize your time (no sketches needed), while adding authenticity to your storyboard.

Use detailed illustrations created with a program such as Adobe, Sketch, or PowerPoint to present storyboards to clients or as a design deliverable.

Remember: a storyboard is meant to tell a story. Don't spend too much time finessing the visuals unless you have to.



Creating a storyboard on sticky notes allows you to be collaborative and rearrange the sequence as you discover new information.

3. Define the basics

Define the persona and the scenario or user story represented. The scenario should be specific and should correspond to a single user path, so that your storyboard doesn't split into multiple directions. For complex, multipath scenarios, maintain a 1-to-1 rule — one storyboard per one path that the user takes. You'll end up with several storyboards, each outlining a different user path. (It's not a flowchart, it's a storyline.)

4. Plan out steps

Start by writing out the steps and connecting them with arrows before going straight to the storyboard template. Next, add the emotional state as an icon to each step, as seen below. This technique will help you start to visualize what each visual frame will include.

5. Create visuals and add captions

While you can use advanced illustration skills to create a beautiful comic-book-quality storyboard, they are not a prerequisite for effective storyboards. It's okay to use stick figures or basic sketches of what you're trying to convey.

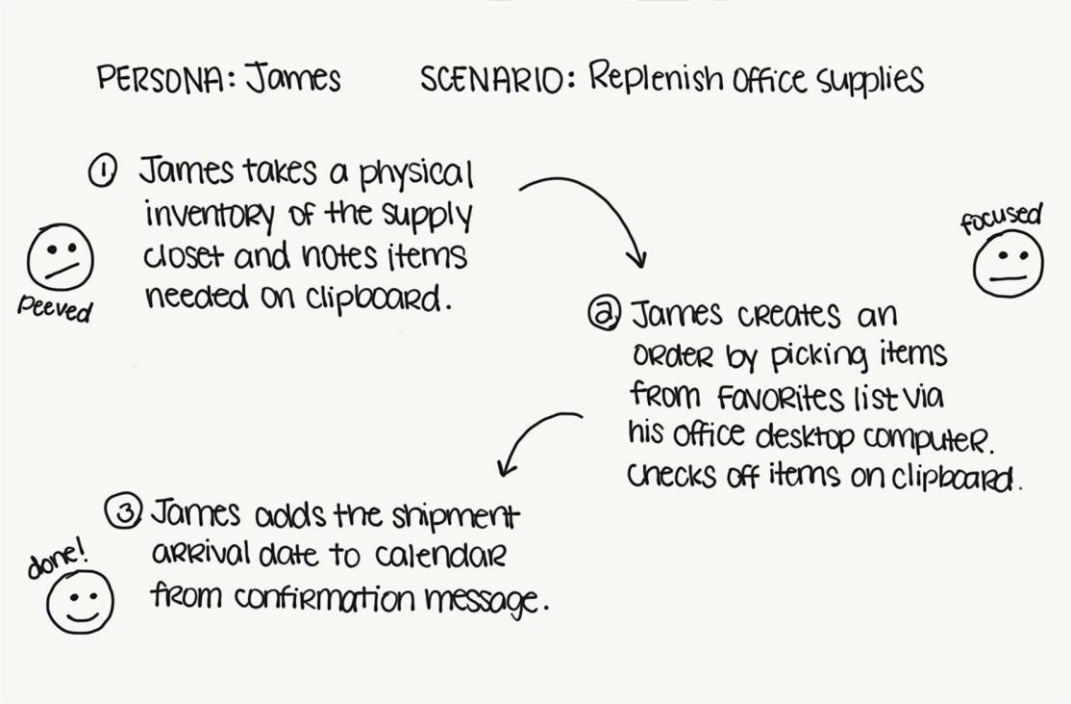
Add captions as bullet points underneath the visuals to describe additional context that is not understood at first glance.

Your storyboard should be in an easy-to-modify format, so that you can make changes in further iterations if necessary. (That is another advantage of low-fidelity sketches or sticky notes.)

While you're not aiming at publication-quality comic-book art, there are so many similarities between storyboards and comic strips that you can get many ideas for visual storytelling and the visualization of emotions, situations, and actions from Scott McCloud's classic book Understanding Comics.

6. Distribute & iterate

Distribute your storyboard to your audience, whether it's your internal team or the stakeholders of the project and ask for feedback. If necessary, iterate over some of these steps to improve the artifact.



If creating the visuals is daunting, start by writing out the steps and determining the user's emotional state for each step.

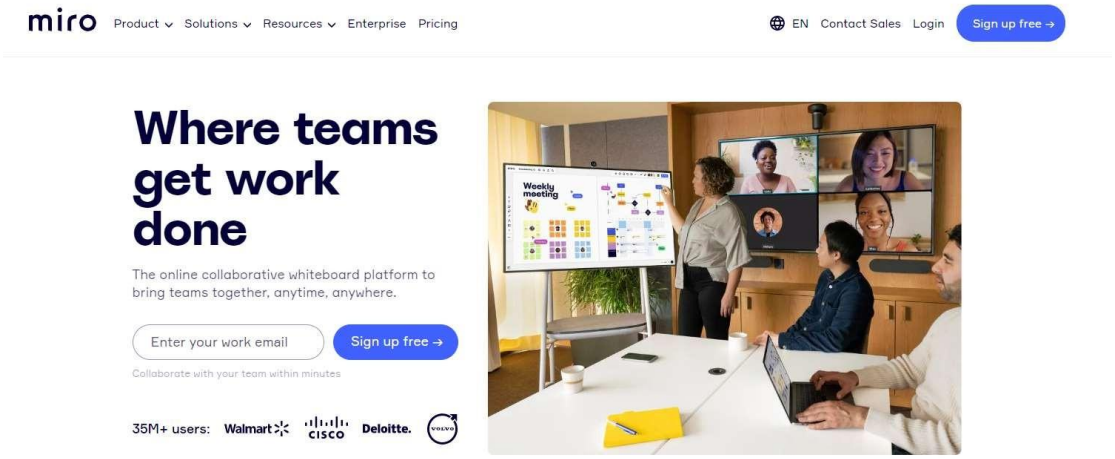
Storyboard tools

Try these different tools for storyboarding in an in-person, remote, or hybrid setting.

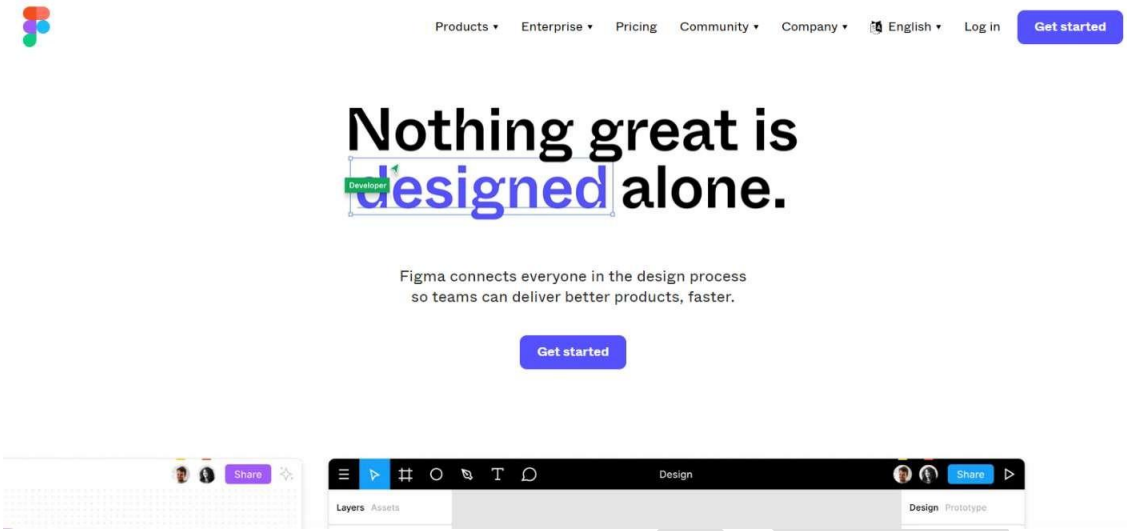
1. *(The old school way) whiteboards, bulletin boards, sticky notes, etc.*



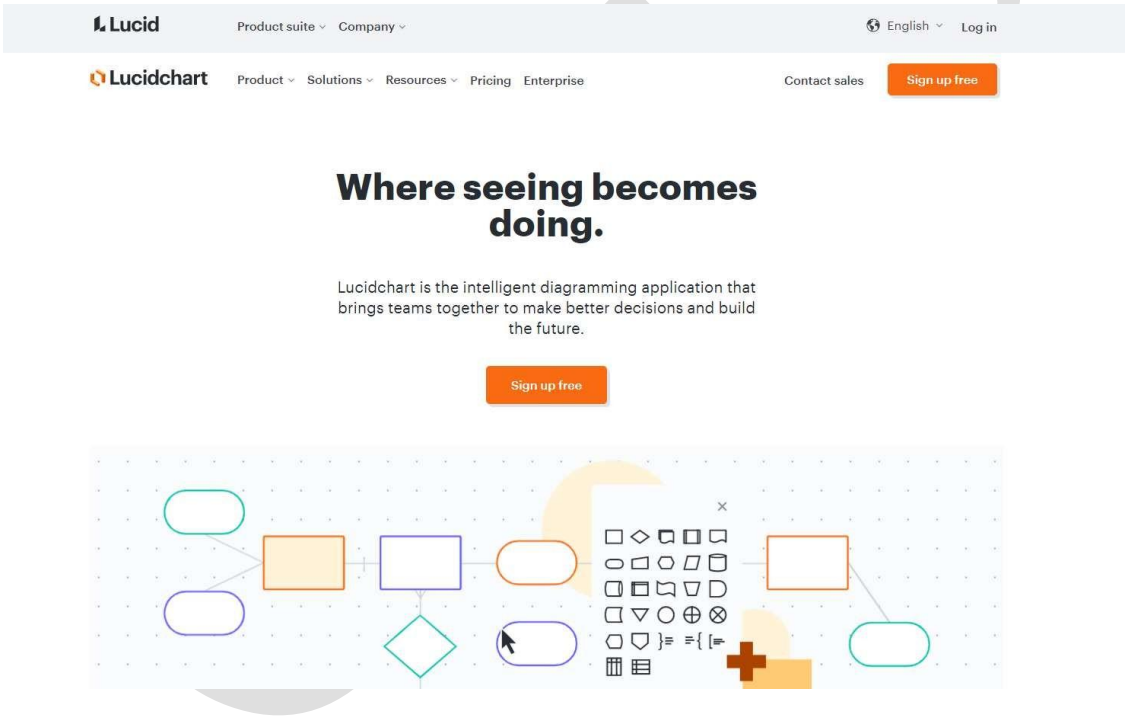
2. Miro



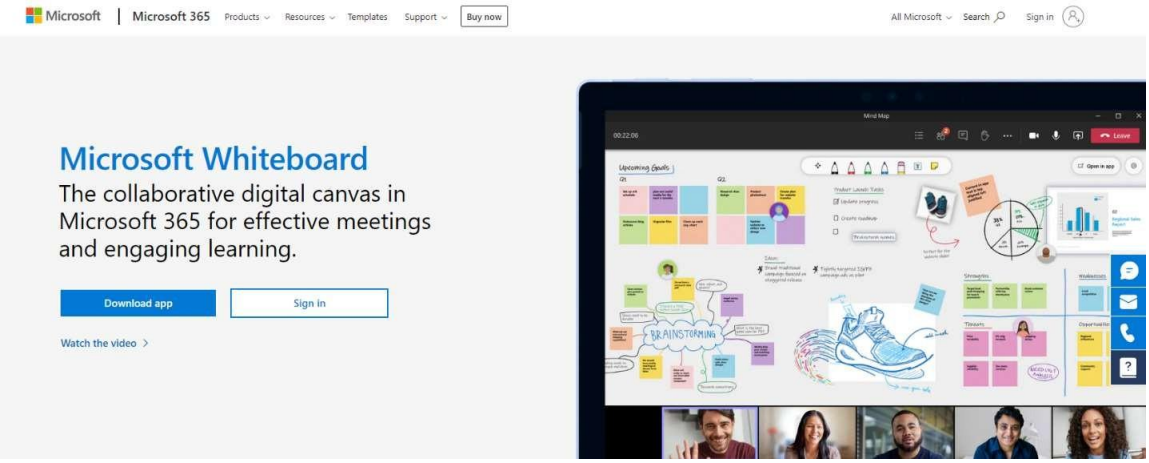
3. Figma



4. LucidChart



5. MicroSoft Whiteboard



BINGO SELECTION

At some point in your ideation sessions, you'll have reached a critical mass of ideas, and it will become unproductive to attempt to keep pushing for more. This is different from the natural creative slumps that teams experience throughout ideation sessions, and means it is a good point to stop and focus on pruning. This is referred to as the 'convergent stage' where ideas are evaluated, compared, ranked, clustered and even ditched in an attempt to pull together a few great ideas to act on. Hang onto those unused ideas, though; they may prove useful in future ideation sessions as stokers or idea triggers. Right now, the aim is spotting potential winners, or combinations of winning attributes, from a number of ideas.

Best practice: How

- The Bingo Selection method inspires participants to divide ideas: The facilitator should encourage the participants to split ideas according to a variety of form factors, such as their potential applications in: A physical prototype, a digital prototype, and an experience prototype.
- Ideation participants decide upon one or two ideas for each of these categories.
- If you're a relatively small team you simply discuss the pros and cons of the chosen ideas.
- If you're a large team you can mix this method with Post-It Voting (also known as Dot Voting):

Post-it Voting

- Write all of the ideas which the participants have chosen on individual Post-its.
- Then you give all participants a number of votes (around three to four should do) to choose and write down their personal favourite ideas.
- Participants vote by using stickers or simply using a marker to make a dot on the ideas they like. This process allows every member to have an equal say in the shortlisted ideas.
- You can also use variations in colour in order to let participants vote on which ideas they like the most or which they dislike the most.

- You can invent other voting attributes when it makes sense.

SIX THINKING HATS

The Six Thinking Hats method was developed by a leader in the field of creative thinking - Edward De Bono, back in 1985. Whatever innovation activity you decide to start; it always has to start with idea generation. Idea generation is a part of the ideation process where your employees are engaged in a fun and creative way to contribute with their ideas. Usually, it results in a vast number of ideas. Identifying the most promising ideas to implement is the much harder part, yet different ideation techniques enable the reduction of ideas at the beginning of the ideation process.

To explore the potential of an idea, De Bono's Six Thinking Hats method can be of great assistance in deciding whether you should or shouldn't proceed with the idea. The method explains the six thinking styles, each represented by a different colored hat. Each hat represents a different perspective. It will help your innovation team to be more productive and focused on the idea by taking the different standpoints.



White Hat (Data, Facts, Figures) Let The Facts Do The Talking

The White thinking hat offers a neutral questioning approach that focuses on the available data and facts regarding the idea. The aim of these questions is to provide objective data, facts and figures about the idea to be discussed. Collect data, facts, relevant and objective information about the problem you are working to solve. You need to be unbiased. Be neutral. Please leave your feelings, opinions, negative thoughts and dirty shoes at the imaginary door.

White Hat Questions:

- What information do we have?
- What are the facts do we have?
- What info is missing?
- How do we get Information?
- What information is important to us?
- Is this valid?
- How do we know?

3 min



Yellow (Optimism) Think Positive, Be Positive

In contrast to the Black Hat, the Yellow Hat questions logical optimistic assessment of an idea. It improves the understanding of the advantages and benefits that certainly helps you identify the value of the idea. Opportunities don't happen, you create them. With this hat, focus on opportunities, the positives, the ideal scenario.

Yellow Hat Questions:

- How and why will this idea work?
- Why do you think this idea is good?
- What are the advantages of the ideas?
- What are the benefits?
- How can we make this happen?
- How will this help our users?
- What do we like about this?
- What will our users like about this?
- How can this reach the maximum potential?



Black (Caution and Risks) Critically Think, Cautiously Compare

Using the Black Hat, your innovation team will be navigated to think about the ideas cautiously and defensively. The aim of this approach is to focus on warnings, risks or cautions and spot difficulties and weak points and eliminate them, along with refinement and improvement of the ideas. Be realistic. Point out the downfalls. Image the absolute worst case scenario. Pessimism is your friend in this hat.

Black Hat Questions:

- What are the potential risks of the idea?
- Why the idea might not work?
- Is this real?
- Can I trust this?
- What are the weaknesses I see?
- Is there proof of errors?
- What are the risks?
- How will this negatively impact us?
- Why won't this work?



Red (Emotions and Feelings) Lead With Emotions

The Red Hat represents feelings and improves the insight into the ideas using your intuition, gut reaction and emotion. When using this hat your team will be able to express their emotions and feelings about the idea. The aim of red hat is not the understanding of the reason behind these feelings, but different emotional reactions regarding the idea. What emotions would your user feel if they interacted with this product/idea? Feel their hypothetical emotional reactions and claim them as your own while wearing this hat.

Red Hat Questions:

- How am I reacting to this?
- How do my users feel about this?
- What is my intuition / gut telling me?
- Does this make me angry/ happy/ excited/ nervous/ annoyed?
- Why am I reacting to this?
- What do you feel about the idea?
- What is your gut reaction toward the idea?



Green (Ideas) Creativity is Contagious

The green hat represents creative thinking with the objective to generate new ideas, possibilities and alternatives out of the existing ideas. Tap into your imagination. Dream up ideas to solve problems. With this hat, you can develop new solutions to previously defined idea issues or to look up for new ideas from a creative perspective. Don't settle for the first or second. Go for as long as you are feeling the creative powers.

Green Hat Questions:

- How can we work this out?
- What is possible?
- What is an alternative way to solve this?
- Can we combine these together?
- What is holding our team back?
- What is something that could be developed from this?
- How can we create value from this?
- What else can we do?



Blue (Control) Get Organized

The blue hat is associated with an overview of the idea and it represents process control. This hat is best worn by the group facilitator. This hat is the first hat & the last hat. The instructional hat that explains the structure and rules of the 6 hats method and prepares the sequence the hats will take. Blue hat keeps the rest of the group on track and wraps up the session when appropriate. The blue hat organizes the next steps, proposes an action plan and sets objectives of the idea. It is also used to summarize and conclude the potential of the idea.

Blue Hat Questions:

Blue Hat Opening:

- Why are we here?
- What do we want to achieve through this?

- How long do we want to wear our hats?

Blue Hat Closing:

- What did we achieve?
- What decisions have we made?
- Where do we go from here?

Applying the Six Thinking Hats (Summary)

Let's understand this through an example. Your organization organizes an ideation activity to collect ideas for new product design. It is beneficial for each participant to individually express his/her thoughts for each idea and to finally consolidate all the ideas. Wearing the White Hat, they will analyse the data they have. They can discuss the facts around a proposed idea such as "do we have required resources", "do we have the technology for this design". Then, Yellow Hat will assist in identifying the advantages and positive impact of the idea. The Black Hat is here to remind us of the potential disadvantages, risks and difficulties that can have a bad impact on sales and production. With Red Hat, you will easily understand the feeling and emotions of the team about the idea, do they like it or not.

Then Green Hat will bring thinking of the idea for a new design from a creative and innovative perspective. This hat helps you bring and connect new ideas to improve the original one. Finally, the Blue Hat will summarize everything, ensuring that the idea has been analysed sufficiently.

MIND MAPPING

Mind mapping is an effective means to take notes and brainstorm essay topics. A mind map involves writing down a central theme and thinking of new and related ideas which radiate out from the centre. By focusing on key ideas written down in your own words and looking for connections between them, you can map knowledge in a way that will help you to better understand and retain information. It is growing diagram composed of keywords, phrases, concepts, facts and figures working outward in all directions starting from a central theme/main idea.

You can use mind mapping for the following:

♣ taking notes in a lecture and listening for the most important points or keywords

♣ showing links and relationships between the main ideas in your subject

♣ brainstorming all the things you already know about an essay question

♣ planning the early stages of an essay by visualising all the aspects of the question

♣ organising your ideas and information by making it accessible on a single page

♣ assignments and essay writing especially in the initial stages, where it is an ideal strategy to use for your 'thinking'

♣ reviewing learning in preparation for a test or examination.

The benefits of mind mapping are in giving you an overview of a large subject/broad topic and allowing you to represent it in a more concise fashion. It encourages you to see the bigger picture and creative pathways

Procedure for mind mapping:

1) Place the central theme/main idea or controlling point in the centre of your page. You may find it easier to place your page on the side, in landscape orientation, which is easier for drawing purposes.

2) Use lines, arrows, speech bubbles, branches and different colours as ways of showing the connection between the central theme/main idea and your ideas which stem from that focus. The relationships are important, as they may form your essay paragraphs.

3) Avoid creating an artistic masterpiece. You should draw quickly without major pauses or editing. Chances are, your first idea was fine and you placed that idea in the direction or on the branch you thought made the most sense. It is important in the initial stages of mind mapping to consider every possibility, even those you may not use.

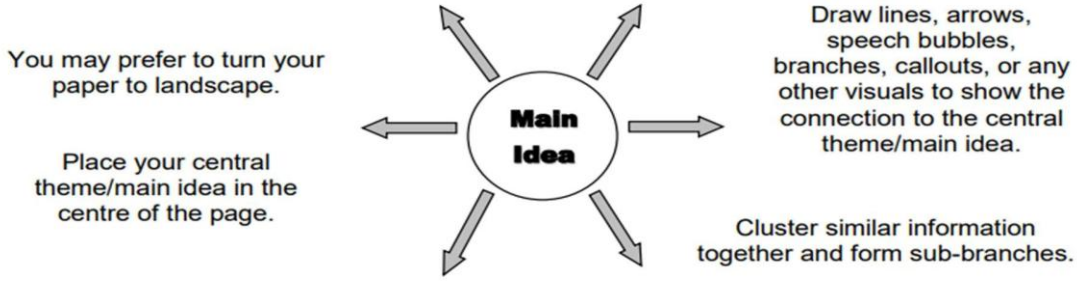
4) Choose different colours to symbolise different things e.g. you may choose blue for something you must incorporate in your paper, black for other good ideas, and red for the things you need to research or check with your tutor/lecturer. Your method is entirely up to you, but try to remain consistent so that you can better reflect on your mind map at a later stage.

5) Leave some space on your page. The reason for this is that you can continue to add to your diagram over a period of time. If A4 sized paper feels too small, you may like to use A3.

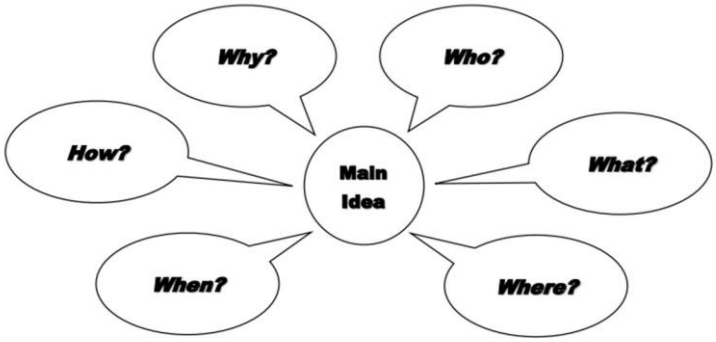
What does a mind map look like?

A standard mind map begins with the following shape

SVB



The following mind map is a basic model with which you can get started on a research topic:





Example: Mind Mapping 'Strategies for Climate Change' Essay Topic

Turning mind map into an academic essay:

Before you can begin to write your essay paper, you must first understand and determine what the 'main' idea of your writing will be. What is the central theme/main idea or controlling point that you are trying to convey to your reader? Whatever the topic, it is always a good idea to think about the issues that surround that topic and mind mapping is an excellent form of brainstorming.

The information that comprises your paragraphs must have a relationship to that controlling point. The decision about what to include in your paragraphs ultimately begins with your ideas. Ideally, each paragraph in an essay should contain one idea or cover one aspect relevant to the central theme on your mind map.

